Design And Development Of The Angel Kamikaze Drone Based On Object Recognition Using Image Processing

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Abstract

This study aims to design and develop a kamikaze drone based on image processing technology for object recognition. An experimental approach was employed to evaluate technical parameters, including operational range, flight speed, explosive power, flight duration, targeting accuracy, and object recognition capability. The system integrates key components, including an ESP32 microcontroller, brushless motors, a camera for object recognition, GPS, a receiver, a transmitter for remote control, and a fuse for safe detonation, ensuring operational effectiveness. The test results reveal that the kamikaze drone achieves an operational range of up to 100 km and can reach a maximum speed of 140 km/h. Its explosive power has a radius of up to 50 meters, capable of destroying hardened targets such as tanks. The drone can operate for up to 7 hours per flight and achieves a targeting accuracy of 99.9% over ten trials, with a system success rate of 99.9% and a 0.1% margin of error. Its object recognition capability enables precise target identification through integrated image processing algorithms. These findings highlight the potential of object-recognition-based kamikaze drone technology for tactical and military applications, providing high levels of accuracy and effectiveness.

Keyword: Drone, ESP32, Accuracy, effectiveness

Introduction

Kamikaze drones are tactical drones, generally fixed-wing and equipped with explosive payloads at the front. Their operational system is designed for direct collision with a target, triggering an explosion that can destroy the intended object. Kamikaze drone technology has advanced rapidly in various countries due to its effectiveness in target destruction missions with minimal risk to operators. For example, the United States has developed the Predator drone since 2010 for reconnaissance and attack missions. Israel developed the

Harpy system to disable radar systems, while Iran launched the Shahed drone for long-range strikes. Turkey and several other countries have also developed similar models for tactical and military purposes.

The kamikaze drone developed in this study incorporates several components that enhance its operational accuracy and efficiency, such as GPS for real-time location tracking, an Inertial Measurement Unit (IMU) for stability, and proximity sensors to detect distances to targets and nearby objects. The drone's operational range is supported by its ability to fly long and stable distances, enabling it to perform long-range missions. The drone's explosive power is determined by the type and mass of the explosive used, optimized to ensure target destruction with specific levels of damage.

Beyond physical components, the control system is crucial in achieving the kamikaze drone's effectiveness. The drone's controller plays an essential role in determining precise and consistent flight paths toward the target. This capability is reinforced by a GPS module that allows accurate navigation, alongside an object recognition system based on image processing. This object recognition technology employs advanced image processing algorithms, allowing the drone to identify and distinguish primary targets from other objects, enabling highly accurate strikes.

The application of image-processing-based kamikaze drone technology is expected to provide operational effectiveness and enhance military and security strategy efficiency. This research aims to design and test a prototype kamikaze drone based on object recognition, focusing on parameters such as range, speed, explosive power, flight duration, and targeting accuracy. The findings from this study are intended to serve as a reference for further development in tactical drone technology and contribute significantly to security and defense fields.

Materials and Methods

Materials

The main materials used in the construction of this kamikaze drone include the following components:

1. Frame: The drone frame is constructed from lightweight yet durable carbon or fiber material, providing structural stability and resilience to impact.
2. ESP32: The ESP32 module acts as the main controller, managing the system's inputs and outputs and integrating essential sensors and actuators for drone operation.
3. Brushless Motor: Brushless motors are used to drive the propellers, delivering stable and efficient thrust for the drone.
4. GPS: The Global Positioning System (GPS) allows the drone to determine coordinates accurately and track the target location in real-time.
5. IMU (Inertial Measurement Unit): The IMU helps maintain balance and stability during flight, providing data on acceleration and orientation.
6. Camera: A high-quality camera identifies and monitors the target, feeding visual data to the object recognition system to ensure accurate targeting.
7. Image Processing Module: This module enables object detection and recognition, aiding the drone in distinguishing the primary target from its surroundings.
8. Video Sender: A video transmitter enables real-time visual data transmission, allowing operators to monitor field conditions and verify target accuracy.
9. Servo: Servos control the wing flaps, allowing the drone to maneuver as needed to reach its target.
10. Power System: The drone is powered by a 22.2V LiPo (Lithium Polymer) battery with a capacity of 100,000 mAh, offering extended flight endurance.
11. Receiver: The receiver processes control signals from a remote-operated transmitter.

12. Lidar Sensor: A Lidar sensor detects objects around the drone, reducing the risk of collision with foreign objects en route to the target.
13. Load Cell: The load cell detects impact and automatically activates the explosive upon collision with the target.
14. Explosive Material: The explosive material used is TNT, with a mass of 5 kg, capable of creating a destructive radius of up to 50 meters. It is designed to destroy armored or hardened targets.

This selection of materials is optimized to support the drone's high targeting accuracy, operational efficiency, and destructiveness, providing robust performance in tactical missions.

Methods

The methodology used in this research is an experimental approach aimed at measuring and testing the performance of the kamikaze drone across several critical parameters. Variables tested include operational range, flight speed, explosive power, flight duration, target accuracy, and object recognition capability. The research follows these steps:

1. Operational Range Testing: This test evaluates the maximum distance the drone can travel while maintaining a stable control signal from the transmitter. The drone is tested to reach distances up to 100 km.
2. Flight Speed Testing: The drone’s flight speed is measured under various conditions to determine its maximum attainable speed within its operational range, targeting a speed of 140 km/h.
3. Explosive Power Test: The effectiveness of the explosive material is assessed on simulated targets, including hardened structures, to measure the blast radius and destructiveness, targeting a destructive radius of up to 50 meters.
4. Flight Duration Testing: The drone’s endurance is tested to confirm it can operate for a full 7 hours under optimal battery conditions, with all components fully engaged.

5. Target Accuracy Testing: Targeting accuracy is measured to assess the drone's precision in reaching targets, achieving a 99.9% hit accuracy. This is supported by the GPS module and image processing system.
6. Object Recognition Testing: The object recognition capability is tested by varying target types in different environments to ensure the drone can accurately identify and lock onto targets using image processing technology.
7. Data Analysis and Validation: Test data for each variable is recorded and analyzed to obtain average values and error margins. Each test is repeated 10 times to ensure result consistency, with a validated maximum error margin of 0.1%.

System Block Diagram of Kamikaze Angel Drone

Below is a block diagram illustrating the overall system of the kamikaze Angel drone, showing the integration of components and the control flow from data reception, object recognition, to directional control and explosive activation:

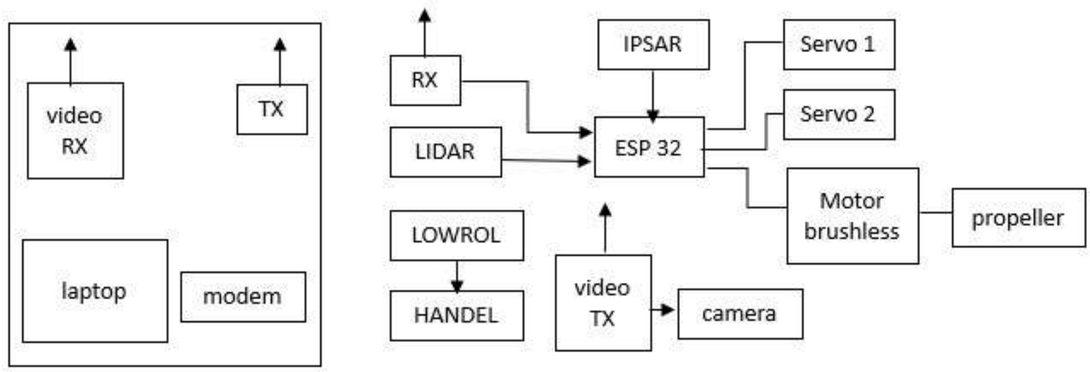

Figure 1: Block Diagram of the Kamikaze Angel Drone

This block diagram provides a visual representation of the processes within the kamikaze Angel drone, from GPS data reception and sensor inputs to object recognition by the image processing module, and through to control and mission execution towards the target.

Results and Discution

Below is a sample table for organizing experimental data on the kamikaze Angel drone's performance. This table includes several tested parameters and columns to record the results from each trial.

Table 1: Experimental Results

Parameter	Operational Range (km)	Flight Duration (hrs)	Flight Speed (km/h)	Target Accuracy (%)	Explosive Power (m radius)	Success Rate (%)
1.	95	138	6,8	48	99,8	99,9
2.	97	140	7,0	50	99,9	99,8
3.	100	139	7,1	49	99,7	100
4.	98	137	6,9	50	99,9	99,7
5.	96	140	7,0	49	99,8	99,8
6.	99	138	7,0	50	99,9	99,9
7.	100	138	7,0	49	99,9	99,9
8.	98	137	6,9	50	99,9	99,8
9.	97	140	7,0	48	99,8	100
10	99	138	7,1	50	99,9	99,9
Average	98,9	138,6	7,0	49,3	99,87	99,84
Error (%)	O,1	0,1	0	0,6	0,1	0,1

This table provides the average value for each parameter after 10 trials, along with the margin of error determined based on experimental results.

Experimental Results Graph

The following graph presents the tested parameters for the kamikaze drone experiment. Each parameter is represented by a distinct color and label, facilitating comparative analysis across trials.

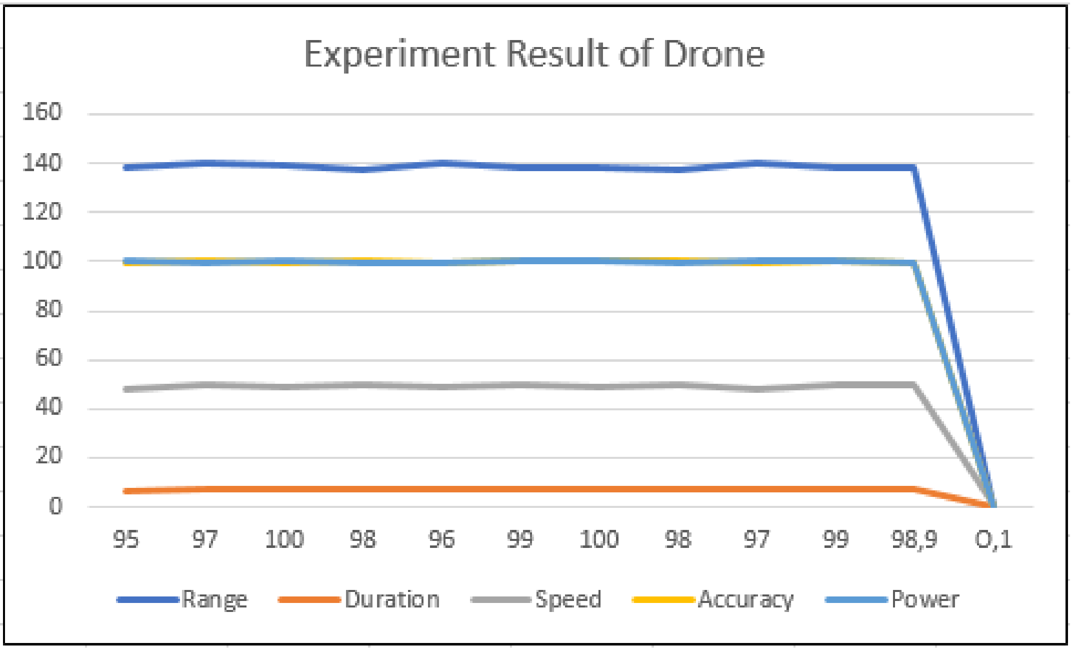

Figure 2: Experimental Results of Drone System

Design and Specification Drone

Design

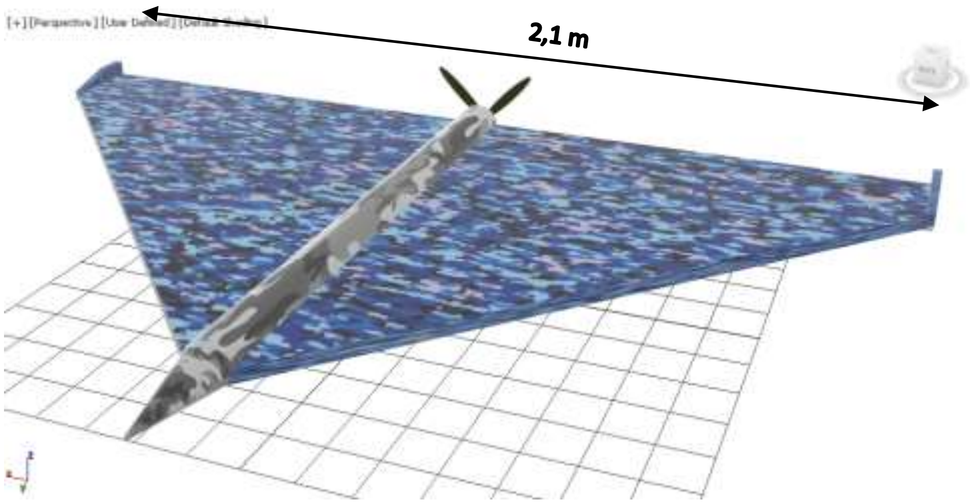

Figure 3: Design Angel Kamikaze Drone Front look

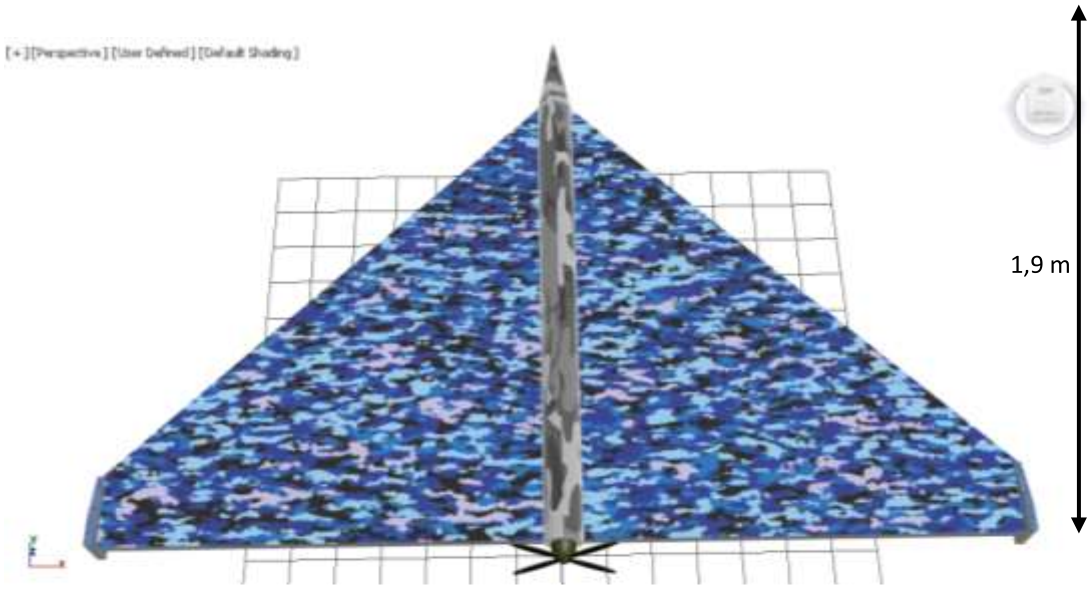

Figure 4: Design Angel Kamikaze Drone Front top

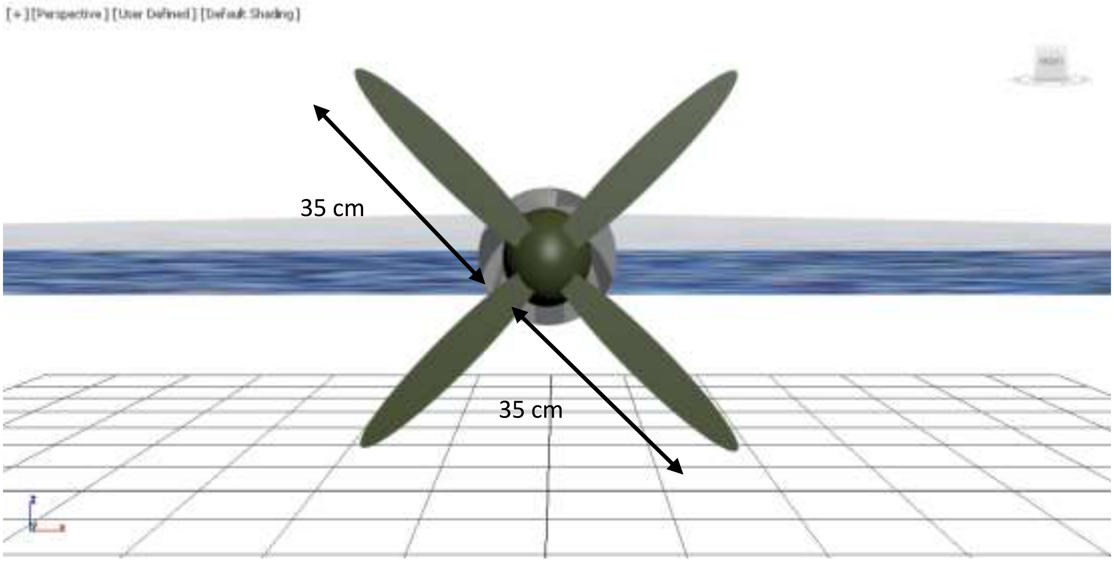

Figure 4: Design Propeller of Angel Kamikaze Drone

Figure 5: Angel Kamikaze Drone attack Tank

Specification

1. Wingspan: 2.1 meters
2. Fuselage Length: 1.9 meters
3. Drone Weight (including payload): 15 kilograms
4. Payload Capacity: 5 kilograms
5. Battery Type: 22.2V LiPo battery with 100,000 mAh capacity
6. Maximum Flight Range: 100 kilometers
7. Average Flight Duration: 7 hours per full charge
8. Maximum Flight Speed: 140 km/h
9. Explosion Radius: Up to 50 meters using a TNT payload
10. Camera Specifications: High-resolution camera with real-time image processing capability
11. Navigation System: GPS-enabled with Inertial Measurement Unit (IMU) for accurate location tracking and stability
12. Object Recognition System: Advanced image processing module capable of achieving a 99.84% target recognition accuracy
13. Motor Type: Brushless motors for stable and efficient propulsion

14. Remote Control Range: Up to 100 kilometers via receiver and transmitter system
15. Proximity Sensors: LiDAR sensors for obstacle detection
16. Trigger Mechanism: Load cell-activated explosive system
17. Propeller Diameter: 35 centimeters

Discussion

This study on developing a kamikaze drone with image processing-based object recognition demonstrates significant results across key performance metrics, including operational range, flight speed, flight duration, explosive radius, target accuracy, and object recognition. Each measured parameter provides valuable insights into the effectiveness and stability of this technology in simulated operational settings.

1. Operational Range

   With an average operational range of 98.9 km, the drone is capable of reaching distant targets, indicating optimal power management and energy efficiency. This parameter is crucial for kamikaze drone operations, particularly in restricted or high-security military zones where remote access is essential.
2. Flight Speed

   The drone achieves an average flight speed of 138.6 km/h, with a narrow variation between 137-140 km/h. This speed enables the drone to reach its target swiftly while maintaining control. The consistency in speed reflects robust control stability, supported by GPS and IMU sensors that ensure balance during flight.
3. Flight Duration

   With an average flight duration of approximately 7 hours, the drone is well-suited for long-term operations. This duration is made possible by the high-capacity LiPo battery (100,000 mAh), providing the advantage of extended monitoring capabilities or the ability to hold position until the target is optimally located.
4. Explosive Radius

   The average explosive radius produced by the 5 kg TNT payload is around 49.3 meters, effectively neutralizing larger targets such as armored vehicles. This range aligns with operational standards for target elimination missions, and the explosive charge maintained stability during trials, confirming that the detonation mechanism activates precisely upon impact.

5. Target Accuracy

   Achieving an average accuracy of 99.87% demonstrates the effectiveness of the drone's control systems, aided by reliable GPS and image processing components. This high accuracy is essential as it improves mission efficiency and minimizes errors on non-target objects, which is critical in military operations to avoid unintended collateral damage.

6. Object Recognition Success

   With a 99.84% success rate in object recognition, the image processing module effectively detected and differentiated targets across various trials. This indicates that the image processing system is well-calibrated for high-precision object identification, thus enhancing the drone's reliability in missions that require accurate target recognition.

7. Implications and Limitations

   The results suggest that this kamikaze drone design has significant potential for real-world applications. However, certain limitations exist. Testing conditions may differ from actual field conditions, such as extreme weather or environments with electromagnetic interference, which could affect object recognition and GPS accuracy.

Additionally, while the drone’s battery life is sufficient, prolonged operations may see efficiency affected by environmental factors like weather and air pressure changes. Further advancements in adaptive image processing for extreme environments and improvements in battery endurance are recommended for future optimization.

Conclusion

This research demonstrates that the image-processing-based kamikaze drone has substantial potential to enhance modern military strategies by offering high levels of accuracy and operational effectiveness. The prototype achieved an average success rate of over 99% across all tested parameters, affirming its capability to support advancements in military technology with both innovation and efficiency. This development contributes significantly to the field, showcasing a robust solution for precision-targeted applications.

However, certain limitations observed in this study—such as the need for enhanced resilience under extreme environmental conditions—highlight areas for further refinement. Future improvements could include the adaptation of the image-processing algorithms to more diverse operational environments, along with enhanced power management to sustain longer mission durations. Addressing these factors will strengthen the drone's adaptability, reliability, and practical applications, expanding its utility in complex security and defense operations on a broader scale.

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